



**SMART INDIA
HACKATHON
2023**

Ministry/Organization Name/Student Innovation:

AICTE, MIC-Student Innovation

PS Code:

SIH1480

Problem Statement Title:

Student Innovation

Team Name:

Team-Kavach

Team Leader Name:

Gopal Agarwal

Institute Code (AISHE):

Institute Name:

Thapar Institute of Engineering and Technology

Theme Name:

Heritage & Culture



THE PROBLEM: HISTORY AT STAKE?

Vandalism

Drawings, writings, symbols, posters?
All ruining the beauty of an
architectural marvel.

Cleanliness and Hygiene

No one appreciates their visit to be
ruined by chips packets and empty
bottles.

Tourist Management

Slow lines? Lost items? Theft?
So much for an organized
and planned tourist experience.

OUR SOLUTION: HERITAGE SHIELD

An **all-encompassing system**, **HeritageShield** offers
complete surveillance and control to the authorities.



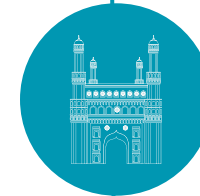
AI Vandalism Surveillance

Cameras enhanced with advanced AI v8
model for vandalism detection and face
recognition machine learning models.



Litter Detection

Machine Learning models trained to
detect littering and tag the defaulters.



Reverse Lost-&Found

1st time ever, Systems to retrieve misplaced
personal items of tourists and detect theft.



QR-Based Ticketing System

Comprehensive QR-based online
ticketing system for convenience and
safety.

Defaulter detected using pose and
face with **D-lib**, & is flagged + fined.

New Vandalism detected using
AI Model trained on Yolo V8.



A Visual Representation of the problem of
Graphic Vandalism



Vandalism Detection & separation from posters/ existing
textures using v8 model to **Avoid false positives**.



An **all-encompassing system**, **HeritageShield** offers
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Technologies Used



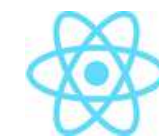
OpenCV



YOLO v8



Flask



React



Py libs



HTML



JS



SHOWSTOPPERS:

Protecting The **Past**,
Securing The **Future**.



Vandalism and Litter Detection

A self designed ML model that can detect vandalism and littering offenders



Financially Practical

Upto **60-70%** reduction (projected) in current running and maintenance costs.



Easy Ticketing and Management

Hassle-free and quick experience for any tourist. skip the lines



Tracking Possesions

Easy location and tracking of any lost items and their owners.

DEPENDENCIES:



Local Server Management

for complete surveillance system



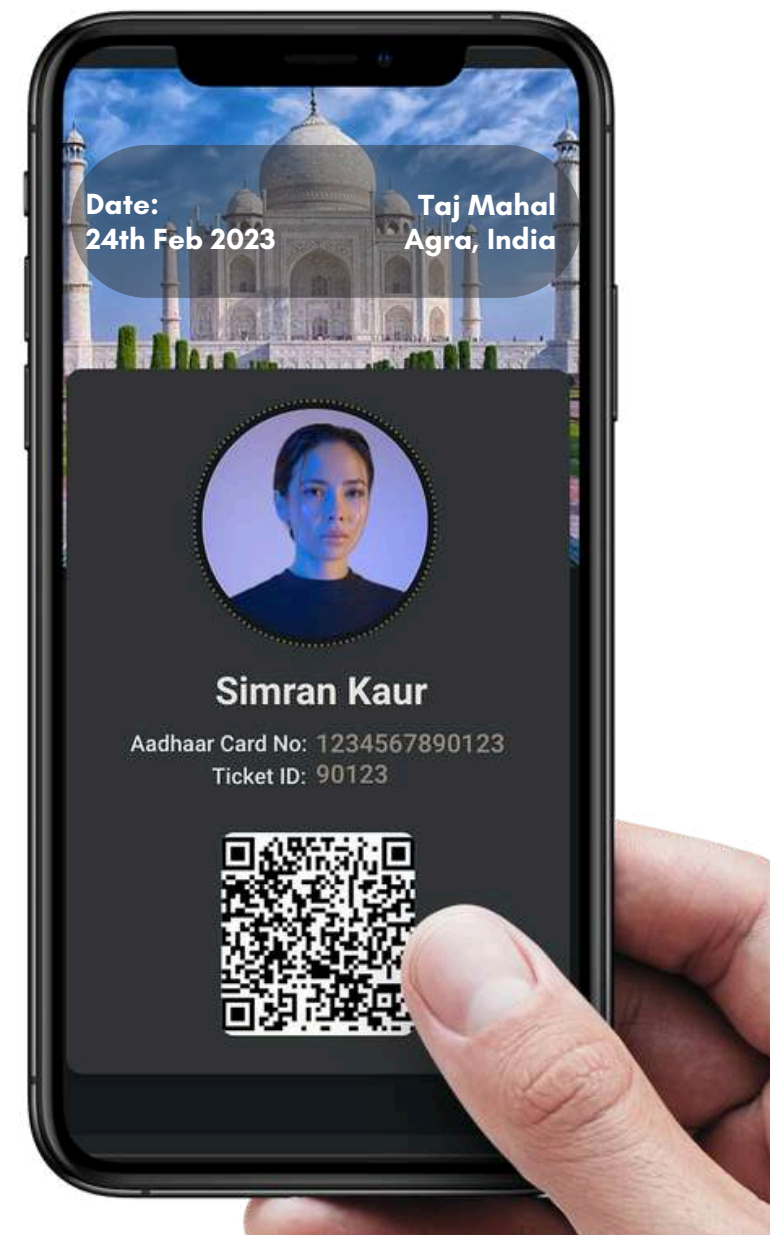
Installation of Extra cameras

Extension of the current camera surveillance system



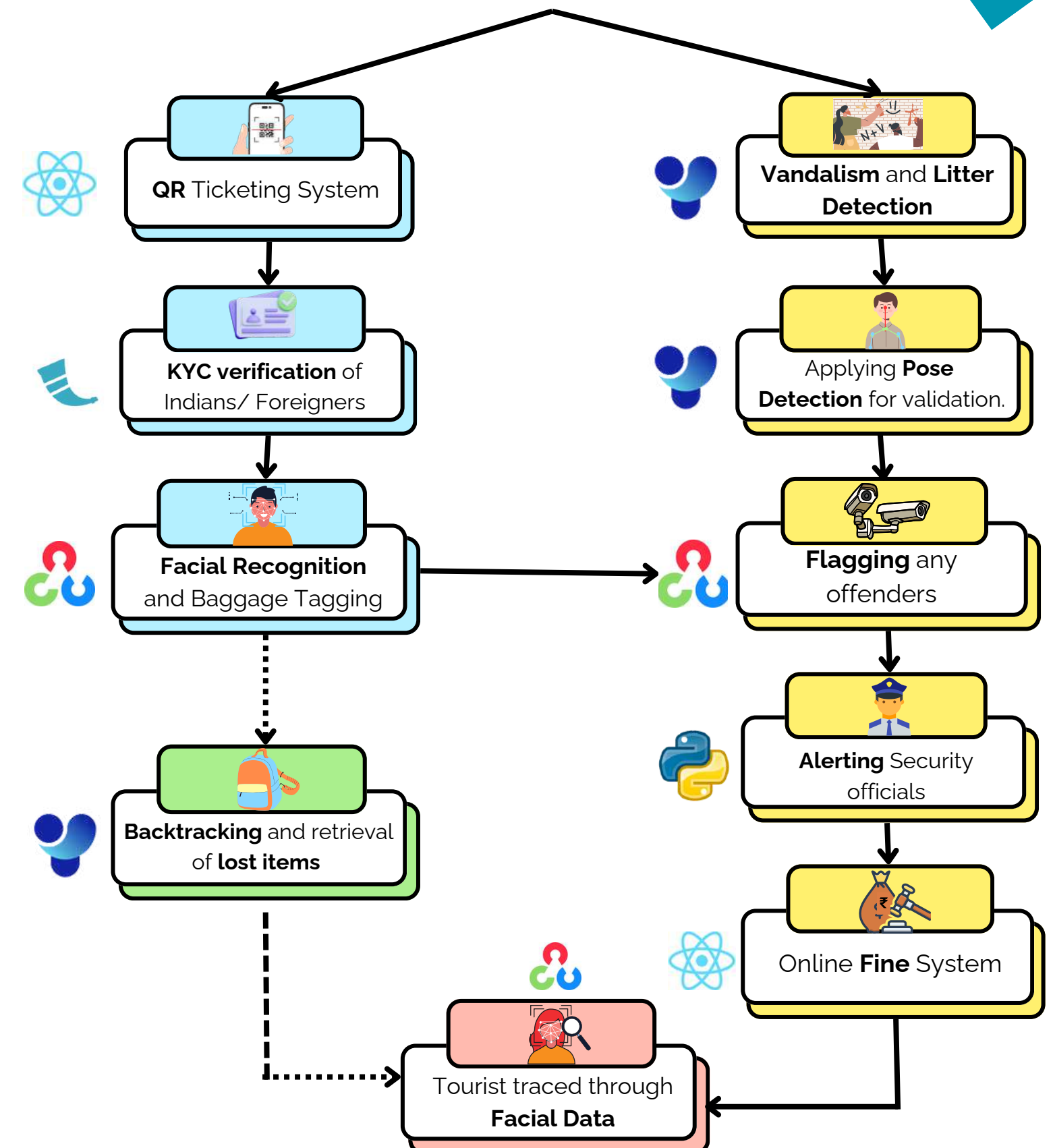
Aadhar Authentication APIs

For KYC during online ticketing for Indian nationality verification



An example of **HeritageShield's** online KYC verified **QR Generated Ticket**

HERITAGESHIELD



Team Member Details

Team Leader Name: Gopal Agarwal		
Branch: B.Tech.	Stream: COE	Year: II
Team Member 1 Name: Shaurya D Verma		
Branch: B.Tech.	Stream: COE	Year: II
Team Member 2 Name: Aaditya Thakur		
Branch: B.Tech.	Stream: COPC	Year: II
Team Member 3 Name: Pranjal Kishor		
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Team Member 4 Name: Gurman Kaur		
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Team Member 5 Name: Manvi Khurana		
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Team Mentor 1 Name: Dr. Sachin Kansal		
Category: Academic	Expertise: Robotics, AI	Domain: Computer Vision